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GIGA Display Shield



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GIGA Display Shield GFX Guide

Learn how to draw shapes, print text and other useful methods with the GFX library.

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The **GIGA Display Shield** is compatible with the **Adafruit_GFX** graphics core library.

To access it, we can use a layer library called **Arduino_GigaDisplay_GFX** designed specifically for the shield. In this guide, we will get started with some of the essential methods of the library, that will allow us to e.g. print values, text, draw shapes.

Goals

In this guide we will cover:

Learn how to draw and print to the display.

How to draw basic shapes (circles, rectangles etc).

How to update values on the screen.

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Arduino GIGA Display
Shield

Arduino IDE

Installation

To install the library, open the Arduino IDE and search for the **Arduino_GigaDisplay_GFX** library in the **Library Manager**.

You can also manually download the library in the official GitHub repository. For any issues with the libraries, please report them here as well.

[Arduino_GigaDisplay_GFX](#)

Basic Example

The [Arduino_GigaDisplay_GFX](#) is a layer library for the [Adafruit_GFX](#) library that's been around for some years now. The full documentation for this library can be found [here](#).

The library provides graphic functions for drawing individual pixels, rectangles, circles, lines and other geometrical shapes. It also provides support for printing numeric values & strings directly on the display.

To use the library, we simply need to create a display object, initialize the library, change something and print it on the display. See this minimal example below:



```
1  #include "Arduino_Giga
2
3  GigaDisplay_GFX display
4
5  void setup() {
6      delay(3000);
7      display.begin(); //i
8
9      display.setCursor(10
10     display.setTextSize(
11     display.print("Hello
12 }
13 void loop(){}

```

The above example will simply print `Hello World!` at the `x` and `y` coordinates (10,10). Simple as that.

Methods Overview

There are several methods available. In this section, we will list out a number of useful ones. To see the full list, check out the Adafruit® documentation of this library in [this page](#).

Coordinates

The coordinates in the GFX library are easy to work with. Most methods have the coordinates as parameters (x,y). To set the cursor (position) on the display, you can utilize the following method:

`setCursor(x, y)` - set the cursor on the display.

Color

To change color of text, pixels,

define the color using the RGB565 format:

```
#define YELLOW 0xFFE0
```

Then use it anywhere in the code for coloring, such as:

```
fillScreen(YELLOW)
```

 - sets the background color for the display.

```
drawPixel(x, y, YELLOW)
```

 - color a single pixel.

```
setTextColor(YELLOW)
```

 - color the text.

```
fillRect(x, y, width, height, YELLOW)
```

 color a rectangle.

Read more about the color format and generate the colors you want at the following page:

[RGB Color Picker](#) (external link)

Text

To display text on the screen, use the following methods:

```
setTextSize(size)
```

 - set the size for the text. Any number between `1` (very small) to `80` (one digit covers the screen) works on this display.

```
setTextColor(color, background_color)
```

- sets color and optionally background color. This is useful when overwriting an existing number (like printing sensor values continuously).

```
print(content)
```

 print something to the screen.

A minimal example for using text

```
1 #include "Arduino_Giga
2
3 GigaDisplay_GFX displa
4
5 #define BLACK 0x0000
6
7 void setup() {
8     delay(3000);
9     display.begin();
10    display.fillScreen(B
11    display.setCursor(10
12    display.setTextSize(
13    display.print("Hello
14 }
15 void loop(){}

```



Pixels & Shapes

To draw pixels and shapes, use the following methods.

`drawPixel(x, y, color)` -
draws a pixel at the
coordinates specified.

`drawRect(x, y, height,
width, color)`
- draws a rectangle at the
coordinates with the height
and width & color specified.

`drawRoundRect(x, y,
height, width, border-
radius, color)`
- rounded rectangle with the
border-radius specified.

`drawTriangle(x0, y0,
x1, y1, x2, y2, color)`
- draws a triangle at the
coordinates specified.

`drawCircle(x, y,
radius, color)`

The above methods will draw the desired shape's outline, but it will not fill it. To do so, simply use `fillRect()`, `fillTriangle()` etc.

A minimal example for drawing geometrical shapes can be seen below:

```
1  #include "Arduino_GigaDisplay_Touch.h"
2
3  GigaDisplay_GFX display;
4
5  #define WHITE 0xffff
6  #define BLACK 0x0000
7
8  void setup() {
9    delay(3000);
10   display.begin();
11   display.fillScreen(WHITE);
12   display.drawTriangle(10, 10, 50, 50, 90, 10);
13   display.drawCircle(10, 10, 5);
14   display.drawRect(10, 10, 40, 40);
15   display.drawRoundRect(10, 10, 40, 40, 10);
16 }
17 void loop() {}
```



GFX & Touch Example

The GFX library can be used together with the [Arduino_GigaDisplay_Touch](#) library. The below example demonstrates how to read a touch

point and trigger a function, using a simple if statement.

The example below is very minimal, and it simply switches on and off a boolean whenever the screen is touched. But it is a good example to get started with if you plan to build your own UI with a touch interface.

```
1  #include "Arduino_GigaDisplay_Touch.h"
2  #include "Arduino_GigaDisplay_GFX.h"
3
4  Arduino_GigaDisplayTouch display;
5  GigaDisplay_GFX displ
6
7  #define WHITE 0xffff
8  #define BLACK 0x0000
9
10 #define screen_size_x 1024
11 #define screen_size_y 768
12
13 int touch_x;
14 int touch_y;
15
16 //increase or decrease sensitivity
17 int trigger_sensitivity;
18 bool switch_1;
19 int counter;
20
21 void setup() {
22     // put your setup code here, to initialize variables
23     Serial.begin(9600);
24     delay(3000);
25     display.begin();
26
27     if (touchDetector.begin()) {
28         Serial.print("Touch interface initialized\n");
29     }
30 }
```

Learn more about the GIGA



Display's touch interface in the [Touch Interface Guide](#).

Summary

In this guide we have covered some basic methods of the GFX library, such as writing text, drawing rectangles and changing colors & text sizes. The GFX library is easy to get started with for beginners, but can also be used to build sophisticated and powerful UIs with advanced usage.

For more tutorials, visit the [documentation page for GIGA Display Shield](#).

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